

CLASSIFICATION OF DATA MINING

Classification of Data Mining Systems

There are several data mining systems available while some are being developed. Data mining systems can be categorized according to various criteria among other classification are the following:

1. **Classification By The Type Of Data Source Mined** : This classification categories data mining systems according to the type of data handled such as spatial data, multimedia data, time series data, text data, World Wide Web etc
2. **Classification By the Data Model Drawn On:** This class categorizes data mining systems based on the data model involved such as relational database, object-oriented database, data warehouse, transactional etc.
3. **Classification By the Kind of Knowledge Discovered:** This classification categorizes data mining systems according to the kind of knowledge discovered or data mining functionalities such as discrimination, characterization, association, clustering etc. Some systems tend be comprehensive systems offering several data mining functionalities together.
4. **Classification By The Mining Techniques Used:** Data mining systems employ and provide different techniques. This class categorizes data mining systems according to the data analysis approach used such as machine learning, neural networks, genetic algorithms, statistics, visualization, database-oriented or data warehouse-oriented. This class also takes into account the degree of user interaction involved in the data mining process such as query- driven systems interactive exploratory systems, or autonomous systems. A comprehensive system would provide a wide variety to data mining technique to fit different situation and options, and offer different degrees of user interaction.

Data Mining Task

Data mining commonly involves four classes of task:

1. Classification

In this task data will be arranged into predefined groups in terms of attributes, one of which is the class. It will find a model for class attribute as a function of the values of other (predictor) attributes, such that previously unseen records can be assigned a class as accurate as possible. For instance an e-mail program might attempt to classify an e-mail as legitimate or spam.

Common algorithms to use are nearest neighbor, Naives Bayes classifier and Neural Network

2. Clustering

Clustering is similar to classification but the groups are not predefined, so the algorithms will try to group similar items together.

3. Regression

This task attempts is similar to find a function which models the data with the least error. A common method is to use Genetic Programming

4. Association Rule Learning

This searches for relationships between variables. For instance, a superstore might gather data of what each customer buys using association rule learning, the superstore can work out what products are frequently bought together that is useful for marketing purposes. This is sometimes called market basket purposes analysis .

Data Mining Issues

Data mining is still in its infancy, although it is rapidly becoming a trend and ubiquitous, it is lately being applied as reliable and scalable tools that outperform older classical statistical methods. Before the study of data mining develops into a conventional, mature and trusted discipline, several issues have to be addressed; some of these issues are discussed in this section, though not exclusive and are not ordered in any way.

Security and Social Issues

Security is a very social issue in data collection either to be shared or intended to be used for strategic decision making. Also, when data are collected from customers profile, user behaviour understanding, students profile, or correlating personal data with other information, huge amounts of sensitive and private information about individuals or companies are collected and stored. Considering the confidential nature of some the data gathered and the potential illegal access to the information it makes the security issue to be very controversial. In addition data mining may disclose new implicit knowledge about individuals or groups that could violate their private policies, especially if there is potential dissemination of the discovered information. Another important that arises from this concern is the appropriate use data mining. As a result of the value of data, database of all kinds of content are regularly sold and because of the advantage that can be attained from implicit knowledge discovered, some vital information could be withheld, which other may be widely distributed and used without control.

Data Quality

Data quality refers to the accuracy and completeness of a data. It is a multifaceted issue that represents one of the biggest challenges for data mining. The quality of data can be affected by the structured and consistency of the data being analyzed. The presence of duplicate records, lack of data standards, timeliness of updates and human error can significantly impact the effectiveness of more complex data mining techniques that are sensitive to subtle differences that may exist in the data.

To improve quality of data, it is somehow necessary to clean the data, which may involve the removal of duplicate records, normalizing the values used to represent information in the database (for example, ensuring that no is represented as a 0 throughout the database, and not sometimes as a O, and sometimes N), accounting for missing data points, removing unrequired data fields, identifying anomalous data points (e.g. an individual whose age is

shown as 215 years), and standardizing data formats (e.g. changing dates so they include DD/MM/YYYY).

User Interface Issues

The knowledge that is discovered by data mining techniques remains useful as long as it is interesting and understandable by the user. Good data visualization eases the interpretation of data mining results and helps users to have better understanding of their needs. A lot of data exploratory analysis tasks are significantly facilitated by the ability to see data in an appropriate visual presentation. There are many visualization ideas and proposals for effective data graphical presentation. However, there is still a lot research needed to accomplish a good visualization tools for large datasets that could be used to display and manipulate mined knowledge.

The major issues related to user interface and visualizations are screen real-estate information rendering, and interaction. The interactivity of data and data mining results is very vital since it provides means for the user to focus and refine the mining tasks, as well as to picture the discovered knowledge from different angles and at different conceptual levels.

Data Source Issues

There are lots of issues related to the data sources; some are practical such as diversity of data types, while others are philosophical like the data excess problem. It is obvious we have an excess of data since we have more data than we can handle and we are still collecting data at an even higher rate. If the spread database management systems has helped increase the gathering of information, the advent of data mining is certainly encouraging more data harvesting. The present practice is to collect as much data as possible now and process it, or try to process it. Our concern is whether we are collecting the right data at the appropriate amount, whether we know what we want to do with it, and whether differentiate between what data is important and what data is insignificant.

Regarding the issues related to data sources, there is the subject of heterogeneous databases and the focus on diverse complex data types. We

store different types of data in a variety of repositories. It is difficult to expect a data mining system to effectively and efficiently achieve good mining result on all kinds of data and sources. Different types of data and sources may require distinct algorithms and methodologies. Presently, there is a focus on relational databases and data warehouses, but other approaches need to be pioneered for other specific complex data types. Therefore the proliferation of heterogeneous data sources, at structural and semantic levels, poses important challenges not only to the database community but also to the data mining community.

Performance Issues

A lot of artificial intelligence and statistical methods exist for data analysis and interpretation; though the methods were not actually designed for the very large data set (i.e. terabytes) data mining is dealing with these days. This has raised the issues of scalability and efficiency of the data mining methods when processing large data. Algorithms with exponential and even medium-order polynomial complexity cannot be of practical use for data mining; instead linear algorithms are usually the standard. Also, sampling can be used for mining instead of the whole dataset.

Other topic that needs to be considered in performance issues includes completeness and choice of samples, incremental updating and parallel programming. Although parallelism can help solve the size problem if the dataset can be subdivided and the results merged later. And incremental updating is very important for merging results from parallel mining, or updating data mining results when new data become available without necessarily re-analyzing the complete dataset.

Interoperability

Data quality is related to the issue of interoperability of different databases and data mining software. *Interoperability refers to the ability of a computer system and data to work with other systems or data using common standards or processes* . It is a very critical part of the larger efforts to improve or enhance interagency collaboration and information sharing through government and homeland security initiatives. In data mining,

interoperability of databases and software is important to enable the search and analysis of multiple databases simultaneously and to help ensure the compatibility of data mining activities of different agencies.

The data mining projects that want to take the advantage of existing legacy databases or trying to initiate first-time collaborative efforts with other agencies or levels of government such as police may experience interoperability problems. Also, as agencies advances with the creation of new databases and information sharing efforts, they will need to address interoperability issues during their planning stages to better ensure the effectiveness of their data mining projects.

Mining Methodology Issues

These issues relates to the different data mining approaches applied and their limitations. Issues such as versatility of the mining approaches, diversity of data available, dimensionality of the domain, the assessment of the knowledge discovered; the exploitation of background knowledge and metadata, the control and handling of noise in data etc. are all examples that would dictate mining methodology choices. For example it is often desirable to have different data mining methods available since different approaches may perform differently depending upon the data at hand.

More so, different approaches may suit and solve user s needs differently. Most algorithms used in data mining assume the data to be noise-free, which of course is a strong assumption. Most of the datasets contain exceptions, invalid or incomplete information which may probably complicate, if not obscure the analysis process and in many cases compromise the accuracy of the results. Consequentially, data preprocessing (i.e. data cleaning and transformation) becomes very essential. Data cleaning preprocessing is often seen as time consuming and frustrating but is one of the most important phase in the knowledge discovery process. Data mining techniques should be able to handle noise in data or incomplete information.

Mission Creep

Mission creep refers to the use of data for purpose other than that for which the data was originally collected. Mission creep is one of the highest risks of data mining as cited by civil libertarians, and represents how control over one's information can be a fragile proposition. This can occur regardless of whether the data was provided voluntarily by the individual or was collected through other means. In fighting terrorism, this takes on an acute sense of urgency, because it creates pressure on both data holders and officials that access the data. To abound an available resource unused may appear to be negligent and data holders may feel obligated to make any information available that could be used to prevent a future attack or track a known terrorist.

Similarly, government officials that are responsible for ensuring the safety of others may be pressurized to use or combine existing databases to identify potential threats. Unlike physical searches, or the detention of individuals, accessing information for purposes other than originally intended may appear to be a victimless or harmless exercise. However, such information use can lead to unintended outcomes and produce misleading results.

One of the primary reasons for misleading results is inaccurate data. All data collection efforts suffer accuracy concerns to some degree. Ensuring the accuracy of information can require costly protocols that may not be cost-effective if the data is not of inherently high economic value (Jeffrey W. Seifert, 2004).

Privacy

Privacy focuses on both actual projects proposed as well as concerns about the potential for data mining applications to be expanded beyond their original purposes (mission creep). As additional information sharing and data mining initiatives have been announced, increased attention has focused on the implications for privacy. For instance, some experts suggest that anti-terrorism data mining applications might be for combating other types of crime as well.

So far there has been little consensus about how data mining should be carried out with several competing points of view being debated. Some

observers suggest that existing laws and regulations regarding privacy protections are adequate and that these initiatives do not pose any threats to privacy. Others argue that not enough is known about how data mining projects will be carried out and that greater oversight is needed. As data mining efforts continue to advance, Congress may consider a variety of questions including, the degree to which government data, whether data sources are being used for purpose other than those for which they were originally designed, and the possible application of the privacy Act to these initiatives (Jeffrey W. Seifert, 2004)

Data Mining Challenges

In this section we shall be describing some of the current research and application challenges for data mining. These challenges are by no means exhaustive but to acquaint the students with the types of problems facing data mining practitioner.

(1). Larger Databases

Databases with hundreds of fields and tables containing millions of records of a multi- gigabyte size are very prevalent, and terabyte (10^{12} bytes) databases are also becoming common. The methods for dealing with large data volume include more efficient algorithms, sampling, and approximation and massively parallel processing.

(2). High Dimensionality

At times there might be no large number of records in the database, but there can be a large number of fields (attributes, variables); so the dimensionality of the problem becomes high. A high-dimensional data set creates problems in terms of increasing the size of the search space for model induction in a combinatorial explosive manner (Usama et al, 1996). Also, it increases the chances of data mining algorithm finding spurious that may not be valid in general. Approaches to this challenge include the methods to reduce the effective dimensionality of the problem and the use of prior knowledge to identify irrelevant variables.

(3). Missing and Noisy Data

This is a very serious challenge especially in business databases. Some important attributes can be missing if the database is not designed with discovery in mind. Possible solutions include the use of more sophisticated statistical strategies to identify hidden variables and dependencies.

(4). Complex Relationship between Fields

Hierarchically structured attributes or values, relations between attributes, and more sophisticated means for representing knowledge about the about of a database will require algorithms that can effectively use such information. Historically, data mining algorithms have been developed for simple attribute-value records, although new technique for deriving relations between variables are being developed. (Usama 1996; Dzeroski 1996; Djoko, Cook, and Holder 1995)

(v). Understanding of the Patterns

In many applications, it is very important to make the discovered patterns more understandable to humans. The possible solutions include graphic representation, rule structuring, natural language generation and technique for visualization of data and knowledge. Rule-refinement strategies can be used to address a related problem. This discovered knowledge might be implicitly or explicitly redundant.

(vi). Over Fitting

When the algorithm searches for the best parameters or one particular model using a limited set of data, it can model not only the general patterns in the data but also any noise specific to the data set resulting in poor performance of the model on test data. Possible solutions include cross-validation, regularization, and other sophisticated statistical strategies (Usama et al,

1996)

(vii). User Interaction and Prior Knowledge

Most of the data mining techniques are not truly interactive and cannot easily incorporate prior knowledge about a problem except in simple ways. Some employ deductive database capabilities to discover knowledge that is used to guide the data mining search while others use Bayesian approaches, which use prior probabilities over data and distributions as one form of encoding prior knowledge.

(viii). Assessing Statistical Significance

Problems similar to that of over fitting occur when the system is searching over many possible models. For example, if a system tests models at the 0.001 significance level, then on average, with purely random data, $N/1000$ of these models will be accepted as significant (Usama, 1996). One way of addressing this problem is to use methods that adjust the test statistic as a function of the search.

(ix). Changing Data and Knowledge

The incessant change in data makes previously discovered patterns invalid; and the variables measured in a given application database can be modified, deleted or augmented with new measurements over time. The possible solutions to these challenges include incremental methods for updating the patterns and treating change as an opportunity for discovery by using it to signal the search for patterns of change (Usama, 1996).

(x). Integration with Other Systems

A stand-alone discovery system might not be very useful, a typical integration issues include integration with a database management system; for example, through a query interface, integration with spreadsheets and visualization tools, and accommodation of real-time sensor readings.

